

Microcontrollers:

Inclusion of security is becoming necessary as more systems become connected

Several microcontroller (MCU) markets are transitioning to higher-performance technologies as designers seek better MCUs for the Internet of Things (IoT). Which high-level trends are defining the future of MCUs? Hear it from the makers of PowerVR graphics architecture as Alexandru Voica, senior technology marketing specialist, Imagination Technologies, speaks with Dilin Anand of EFM.



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Q. What are some innovative SoCs or other chips for new niches that you have found to be exciting?

A. There have been several systems on chips (SoCs) and platforms released recently that are really exciting. I would start with Samsung ARTIK 1, an IoT development board that uses a dual-core MIPS M-class central processing unit (CPU) cluster.

Ingenic M200 is another great example of an innovative SoC built from the ground up for wearable devices. It implements a power-saving hardware architecture where a high-performance MIPS CPU clocked at 1.2GHz tackles most of the heavy lifting, while less-demanding tasks are handled by a secondary low-power 300MHz MIPS CPU.

Finally, Baikal-T1 processor is the first Russian offering for the communications market to use a MIPS P-class Warrior CPU, boasting highly-competitive properties in terms of performance, technology node and compatibility.

Q. What are the most popular features that you have seen in these IoT-targeted chips?

A. Many IoT-focused chips are integrating advanced functionalities to reduce power and manufacturing costs. One example is the emergence of the sensor-fusion hub. Another big trend is the addition of multi-standard radio on-chip, particularly in connected applications. Inclusion of security is becoming necessary as more systems become connected.

Q. Any shift in the way of thinking of chip vendors as they prepare to fully tackle the IoT space?

A. New or established, every semiconductor vendor needs a new paradigm to solve four critical issues in the IoT, namely, design time, differentiation, time-to-revenue and design cost.

In the IoT market, consumers have high expectations; devices need to be affordable, power-efficient and must work out of the box. To address these issues, we have partnered with Taiwan Semiconductor Manufacturing Co. to create multiple IoT sub-systems using MIPS, PowerVR and Enigma that are optimised for multiple process technologies, from 55nm to 10nm. Using these intellectual property (IP) platforms, semiconductor manufacturers can address the widest range of high-growth markets, from

wearables and sensors to mobile, connected cars, cloud processing and beyond.

Q. How would you define the design of an MCU as we move further towards IoT applications?

A. There are a few essential metrics that define the design of any MCU—ease of software development, low power consumption, wide input/output functionality and performance efficiency—and their importance varies with the target application.

Industrial automation and electronic control units markets, for example, require tight and deterministic operations in safety-critical conditions. For automotive infotainment, those requirements are more relaxed in favour of better raw performance needed to process multimedia data. In the case of smartcities and home automation, having a complete, fully-integrated solution (MCU + baseband radio + radio frequency) is extremely important to accelerate time-to-market.

Q. How are the latest chips reducing power consumption, and what changes are driving this?

A. MIPS is a pure reduced-instruction-set-computer architecture that achieves better performance at lower frequencies and in a smaller area by using power-saving instructions, enabling fine-grain control over all parts of the chip. All MIPS CPUs implement fine-grain clocks, gating to reduce dynamic power and voltage/frequency scaling to allow on-the-fly clock-frequency changes.

The multi-threaded technology in MIPS I-class cores combines multiple processing elements in a single processor core, providing performance (and lower area/power) that would otherwise be required of multiple processor cores.

Q. With Xiaomi's plans to launch its own smartphone processor through Leado, there seems to be a bigger focus on custom chips designed by fabless design houses. Your views?

A. Taking the vertically-integrated route can offer substantial advantages, if done right. Companies that are vertically integrated have a very tight grip on many aspects of manufacturing and there is a lot to be gained from the synergies that result from this. ●